# Smarter Healthcare: Assessing Nurse Workload using Wearable Sensors and Machine Learning

Viet-Hoan Bui[1], Duc-Nghia Tran[2], Hoang Trong Minh[3], Ngoc-Anh Dinh[1], Minh La Tuan[1], and To-Hieu Dao[1,*]
[1]*Phenikaa School of Engineering, Phenikaa University, Duong Noi, Hanoi 12116, Vietnam*
[2]*Institute of Information Technology, VAST, Hanoi, Vietnam*
[3]*Posts and Telecommunications Institute of Technology, Hanoi, Vietnam*
Email: nghiatd@ioit.ac.vn, hoangtrongminh@ptit.edu.vn, hieu.daoto@phenikaa-uni.edu.vn

***Abstract*—Nursing care activity recognition is vital for optimizing workflows and reducing documentation burden in elderly care. The SONAR dataset, recently released as a benchmark for nursing activities and recorded with wearable inertial sensors in a clinical environment, poses unique challenges due to noise, class imbalance, and fine-grained activity definitions. This study presents a systematic evaluation of classical ensemble learning methods on SONAR under deployment-oriented constraints for resource-limited microcontrollers. We assess the impact of sensor placement and model configuration on recognition performance. Among the evaluated algorithms, XGBoost consistently outperformed Extra Trees, LightGBM, and Random Forest, achieving 64.01 % accuracy and 61.21 % weighted F1. These findings highlight both the potential and the remaining challenges in developing reliable and resource-efficient nursing activity recognition systems.**

***Index Terms*—ensemble learning, healthcare monitoring, human activity recognition, nursing care activity recognition**

## I. Introduction

Rapid aging of populations worldwide is exerting unprecedented pressure on healthcare and long-term care systems [1]. According to the World Health Organization, by 2030 one in six people globally will be aged 60 years or older, and the number of older adults is projected to exceed 2.1 billion by 2050 [2], [3]. Countries such as Japan, Germany, and the United Kingdom are already facing severe workforce shortages in caregiving roles, with facility managers consistently reporting difficulties in recruiting and retaining sufficient staff [2]. This demographic shift increases the demand for nursing services and intensifies the strain on an already limited labor force, threatening the sustainability of care delivery.

Within care facilities, nurses and professional caregivers must perform a broad spectrum of essential but highly time-consuming tasks, among which documentation is particularly burdensome [4], [5]. It has been reported that nurses spend between 26.2 % and 41 % of their working time on documentation, making it one of the largest contributors to perceived workload [4], [6], [7]. Excessive documentation requirements reduce the time available for direct patient interaction, thereby exacerbating stress and contributing to burnout. Traditional workflow assessment methods, such as time–motion studies, rely on manual observation and note-taking, which are labor-intensive, costly, and difficult to scale [2], [8]–[10]. As a result, documentation-related workload decreases care efficiency and limits opportunities to provide high-quality, personalized care.

To address these challenges, nursing care activity recognition (NCAR) has emerged as a promising technological approach [2]. As a specialized domain within human activity recognition (HAR), NCAR leverages wearable sensors and machine learning to automatically detect, classify, and analyze nursing activities [2], [11], [12]. By enabling automated monitoring and visualization of workflows, NCAR can support process optimization, alleviate the documentation burden, and contribute to workforce efficiency [2], [4], [13]. Recent reviews indicate that NCAR has the potential to mitigate labor shortages and improve quality of care by standardizing work practices and enabling data-driven insights into caregiver workloads [2], [4], [14].

Despite its promise, NCAR faces persistent challenges that hinder robust deployment. Unlike general HAR, nursing tasks are more diverse and complex [4], [14], spanning from simple mobility assistance to intricate patient-care procedures, which makes it difficult to capture subtle nuances in sensor signals [14], [15]. Nursing activities are highly context-dependent, influenced by environmental conditions, patient status, and individualized care plans [14], [16]. NCAR also exhibits pronounced intra-class variability, as nurses often perform tasks on behalf of patients rather than for themselves [14]. Data collection in clinical environments further introduces missing labels, class imbalance, non-uniform sampling rates, redundant or noisy signals, and limited annotation capacity [2], [4], [14]. Consequently, many state-of-the-art HAR techniques degrade when transferred from laboratory to clinical settings [2], underscoring the need for systematic evaluation of algorithms and deployment strategies that are resilient to these domain-specific challenges.

In addition to retrospective analyses, NCAR systems must also operate in real time. Real-time monitoring transforms NCAR from a purely analytical tool into a proactive assistant for care delivery: it enables dynamic workload management by revealing task distribution during shifts, allows supervisors to reallocate staff when one caregiver is overloaded, and supports timely coordination across the care team. Moreover, automatic documentation can be generated on the fly, directly reducing the substantial time nurses spend on charting.

Delivering such real-time functionality in practice calls for inference directly on wearable devices or small gateways, where continuous connectivity to cloud infrastructure cannot

be guaranteed [17]–[21]. Deployment on resource-constrained microcontrollers (typically 128–256 kB RAM and about 1 MB Flash [22]) provides low-latency operation, scalability across multiple caregivers, and enhanced privacy by avoiding transmission of raw sensor signals. These benefits, however, impose strict memory and computational budgets, motivating compact models that balance accuracy and efficiency under tight size limits.

In response to these challenges, this study offers three contributions. First, we conduct a systematic evaluation of classical ensemble learning methods on the SONAR dataset under deployment-oriented constraints that reflect the memory limitations of low-power microcontrollers (LPMC). Second, we examine a wide range of sensor configurations and highlight practical trade-offs between recognition performance and hardware complexity to inform resource-aware system design. Third, we apply a rigorous nested cross-validation protocol within a 1 MB model-size constraint to establish a reproducible benchmark for NCAR. Together, these contributions clarify the strengths and limitations of compact ensemble models and provide guidance for developing reliable, resource-efficient NCAR systems.

## II. Materials and Methods

### A. Dataset

The experiments used the SONAR (Sensor-Oriented Nursing Activity Recognition) dataset [4], a publicly available benchmark collected in a operational elderly care facility where 14 nurses wore five Xsens DOT v2 IMUs sampling at 60 Hz. As illustrated in Fig. 1, sensors were placed on the right wrist (RW), left wrist (LW), pelvis (ST), right ankle (RF), and left ankle (LF). The dataset comprises 254 recording files with a total of 5,673 recordings covering 23 nursing activities commonly performed in assisted living. Each recording provides multichannel CSV streams containing 70 sensor-derived channels plus timestamp and label, accumulating more than 13 million samples. This rich set of wearable inertial data enables fine-grained analysis of complex nursing tasks in realistic clinical conditions.

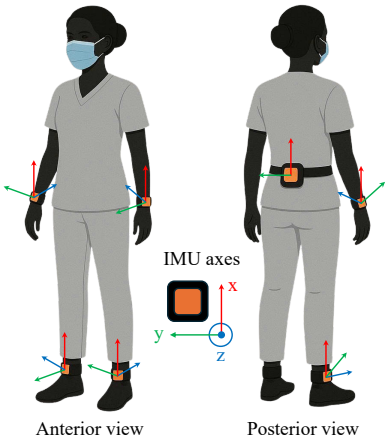


Fig. 1. Body locations of the five IMU sensors used in SONAR

TABLE I
Segmentation Statistics of SONAR Activities

| Activity | Code | Window | Proportion (%) |
|---|---|---|---|
| Null activity | NA | 5,378 | 33.54 |
| Change clothes | CC | 1,888 | 12.48 |
| Serve food | SF | 1,255 | 8.29 |
| Wash in bed | WB | 867 | 5.73 |
| Kitchen preparation | KP | 709 | 4.69 |
| Clean up | CU | 697 | 4.61 |
| Make bed | MB | 814 | 5.38 |
| Deliver food | DF | 486 | 3.21 |
| Wash at sink | WS | 594 | 3.93 |
| Push wheelchair | PW | 469 | 3.10 |
| Put food on plate | PF | 394 | 2.60 |
| Collect dishes | CD | 292 | 1.93 |
| Prepare bath | PB | 307 | 2.03 |
| Wheelchair transfer | WT | 157 | 1.04 |
| Put medication | PM | 236 | 1.56 |
| Pour drinks | PD | 138 | 0.91 |
| Dental care | DC | 133 | 0.88 |
| Documentation | DM | 128 | 0.85 |
| Comb hair | CH | 72 | 0.48 |
| Wipe up | WU | 49 | 0.32 |
| Put accessories | PA | 29 | 0.19 |
| Blow-dry | BD | 31 | 0.20 |
| Wash hair | WH | 8 | 0.05 |

### B. Data Preprocessing and Windowing

The raw SONAR dataset contains a notable proportion of missing values, with an initial duration of 61.66 hours reduced to 49.05 hours after removing all frames containing NaNs, meaning that approximately 20.45 % of the sensor data was affected. Missingness varied considerably across recordings—most exhibited less than 1 % missing values, whereas one recording reached 11.7 %—making imputation unreliable due to the need to reconstruct long signal gaps that could distort temporal dynamics. Given the fine-grained nature of nursing activities, this study adopted a conservative strategy in which any segment containing NaN values in any channel was discarded, yielding a clean dataset without missing values or duplicate timestamps for consistent downstream processing. Following this cleaning step, continuous sensor streams were segmented into fixed-length windows of 600 samples (10 seconds at 60 Hz) without overlap, a choice supported by prior work demonstrating that longer windows capture richer temporal information for complex activities [4]. The resulting segmentation statistics for all 23 activities are summarized in TABLE I.

### C. Feature Extraction

Feature extraction plays a central role in classical machine learning pipelines by transforming raw sensor measurements into discriminative representations suitable for lightweight deployment. In this study, all 14 measurements provided by each IMU—including orientation (quaternions), angular velocity, acceleration, and magnetic field—were used to compute 11 statistical and temporal-domain features for every channel: Hjorth mobility, Hjorth complexity, interquartile range (IQR), maximum, minimum, mean, median, median absolute successive difference (MASD), range, root mean square (RMS),

and standard deviation (SD). These descriptors capture central tendency, variability, amplitude distribution, and signal complexity, with Hjorth parameters being widely used in biomedical time-series analysis and robust statistics such as IQR and MASD offering resilience to noise. Their effectiveness in wearable-sensor activity recognition has been demonstrated in prior studies conducted in operational settings [17]–[21].

### *D. Recognition Models and Validation Strategies*

This study evaluates four classical ensemble classifiers—Extra Trees, LightGBM, Random Forest, and XGBoost—selected for their robustness to class imbalance, noisy sensor measurements collected in clinical environments, and heterogeneous sensor placements in SONAR. Such characteristics may make deep learning approaches less suitable for this dataset, a trend also reflected in the NCAR Challenge, where participating teams relied on ensemble-based CML models rather than DL methods [2]. An additional practical motivation is that these models can be exported to compact C/C++ code using tools such as `m2cgen` and `micromlgen`, enabling deployment on LPMC with strict memory budgets [22]. Empirical memory profiling revealed that feasible hyperparameter ranges were constrained by the 1 MB model-size limit, with the maximum viable combinations reaching $(5, 12)$ trees and depth for Extra Trees, $(9, 10)$ for LightGBM, $(9, 11)$ for Random Forest, and $(29, 3)$ for XGBoost.

To evaluate performance under these constraints, two complementary cross-validation (CV) strategies were adopted. Vanilla cross-validation (VCV) used stratified 5-fold CV to identify hyperparameters that satisfied the model-size requirement, effectively pruning the search space. Because tuning and evaluation occur on the same folds, VCV may overestimate generalization. Therefore, nested cross-validation (NCV) was subsequently applied to obtain an unbiased performance estimate: the inner loop selected hyperparameters within the pruned space, while the outer loop assessed generalization stability across folds. Model performance was reported using accuracy and weighted F1-score expressed as Mean $\pm$ SD, with the latter chosen for consistency with the original SONAR study [4] and its suitability for imbalanced multi-class datasets.

## III. Result and Discussion

### *A. Impact of Sensor Configurations*

For the single-sensor setting (TABLE II), the RW sensor yielded the best performance, achieving an F1-score of 49.00 $\pm$ 0.49 % with XGBoost. This result can be intuitively explained by the predominance of right-hand dominance among participants, as fine-grained manipulative actions performed with the dominant hand provide richer kinematic cues than other body parts. The LW and ST followed closely, but were outperformed by RW.

When two sensors were combined (TABLE III), the inclusion of the ST sensor markedly improved performance. In particular, the (RW, ST) pair reached 55.93 $\pm$ 0.65 % with XGBoost, outperforming other two-sensor configurations. This highlights the complementary role of ST, which captures

TABLE II
5-fold VCV Results with a Single Sensor

| Sensors | Model | Accuracy | $F1_{weighted}$ |
|---|---|---|---|
| LW | Extra Trees | 46.45 ± 0.43 | 37.97 ± 0.60 |
| | LightGBM | 50.62 ± 0.69 | 44.57 ± 0.74 |
| | Random Forest | 47.50 ± 0.67 | 41.47 ± 0.52 |
| | XGBoost | **53.51 ± 1.03** | **47.70 ± 0.81** |
| RW | Extra Trees | 45.21 ± 0.62 | 38.16 ± 0.40 |
| | LightGBM | 50.14 ± 0.73 | 44.38 ± 0.51 |
| | Random Forest | 48.07 ± 0.16 | 41.42 ± 0.16 |
| | XGBoost | ***53.92 ± 0.50*** | ***49.00 ± 0.49*** |
| ST | Extra Trees | 45.97 ± 1.01 | 38.10 ± 1.25 |
| | LightGBM | 50.88 ± 0.64 | 44.81 ± 0.59 |
| | Random Forest | 47.46 ± 0.66 | 41.47 ± 0.64 |
| | XGBoost | **53.12 ± 0.41** | **47.43 ± 0.42** |
| LF | Extra Trees | 45.03 ± 0.53 | 35.30 ± 0.81 |
| | LightGBM | 49.37 ± 0.42 | 42.74 ± 0.50 |
| | Random Forest | 46.33 ± 0.67 | 38.56 ± 0.68 |
| | XGBoost | **51.20 ± 0.50** | **44.70 ± 0.50** |
| RF | Extra Trees | 43.72 ± 0.70 | 35.70 ± 1.06 |
| | LightGBM | 48.98 ± 0.88 | 41.97 ± 0.87 |
| | Random Forest | 45.80 ± 0.52 | 38.93 ± 0.57 |
| | XGBoost | **51.66 ± 0.33** | **45.01 ± 0.40** |

TABLE III
5-fold VCV Results with Two-Sensor Combinations

| Sensors | Model | Accuracy | $F1_{weighted}$ |
|---|---|---|---|
| LW, RW | Extra Trees | 49.08 ± 0.58 | 41.90 ± 0.69 |
| | LightGBM | 54.21 ± 0.83 | 49.52 ± 0.89 |
| | Random Forest | 52.54 ± 0.86 | 45.44 ± 0.69 |
| | XGBoost | **58.71 ± 0.87** | **54.77 ± 0.82** |
| LW, ST | Extra Trees | 50.46 ± 0.62 | 42.71 ± 0.47 |
| | LightGBM | 56.38 ± 0.51 | 51.82 ± 0.46 |
| | Random Forest | 53.84 ± 0.35 | 47.23 ± 0.32 |
| | XGBoost | **59.60 ± 0.98** | **55.50 ± 0.91** |
| LW, LF | Extra Trees | 49.89 ± 0.39 | 41.87 ± 0.50 |
| | LightGBM | 55.44 ± 0.70 | 50.59 ± 0.83 |
| | Random Forest | 52.63 ± 0.67 | 46.42 ± 0.50 |
| | XGBoost | **59.35 ± 0.68** | **55.09 ± 0.63** |
| LW, RF | Extra Trees | 49.85 ± 0.35 | 42.00 ± 0.32 |
| | LightGBM | 55.91 ± 0.64 | 51.21 ± 0.82 |
| | Random Forest | 53.98 ± 1.04 | 46.72 ± 1.06 |
| | XGBoost | **59.14 ± 1.24** | **54.81 ± 1.04** |
| RW, ST | Extra Trees | 48.41 ± 0.55 | 41.66 ± 0.54 |
| | LightGBM | 55.81 ± 0.41 | 51.35 ± 0.51 |
| | Random Forest | 51.30 ± 0.80 | 46.21 ± 1.05 |
| | XGBoost | ***59.63 ± 0.53*** | ***55.93 ± 0.65*** |
| RW, LF | Extra Trees | 48.66 ± 0.56 | 40.35 ± 0.46 |
| | LightGBM | 55.59 ± 0.86 | 51.06 ± 0.87 |
| | Random Forest | 51.19 ± 0.52 | 44.81 ± 0.48 |
| | XGBoost | **58.67 ± 0.93** | **54.90 ± 0.91** |
| RW, RF | Extra Trees | 49.20 ± 0.65 | 41.41 ± 0.71 |
| | LightGBM | 55.06 ± 0.82 | 50.44 ± 0.87 |
| | Random Forest | 52.83 ± 0.57 | 45.21 ± 0.68 |
| | XGBoost | **59.09 ± 0.45** | **55.30 ± 0.47** |
| ST, LF | Extra Trees | 45.13 ± 0.76 | 45.13 ± 0.76 |
| | LightGBM | 52.52 ± 0.54 | 47.31 ± 0.56 |
| | Random Forest | 49.18 ± 0.59 | 41.94 ± 0.81 |
| | XGBoost | **55.18 ± 0.75** | **50.41 ± 0.89** |
| ST, RF | Extra Trees | 45.47 ± 0.18 | 38.13 ± 0.49 |
| | LightGBM | 52.62 ± 0.61 | 47.33 ± 0.55 |
| | Random Forest | 48.31 ± 0.83 | 42.63 ± 0.86 |
| | XGBoost | **55.83 ± 0.41** | **50.93 ± 0.53** |
| LF, RF | Extra Trees | 42.28 ± 0.37 | 36.91 ± 0.46 |
| | LightGBM | 50.00 ± 0.64 | 44.09 ± 0.71 |
| | Random Forest | 48.03 ± 0.56 | 40.19 ± 0.57 |
| | XGBoost | **52.96 ± 0.49** | **47.05 ± 0.47** |

TABLE IV
5-FOLD VCV RESULTS WITH THREE-SENSOR COMBINATIONS

| Sensors | Model | Accuracy | $F1_{weighted}$ |
|---|---|---|---|
| LW, RW, ST | Extra Trees | 50.41 ± 0.64 | 44.21 ± 0.64 |
| | LightGBM | 58.06 ± 0.54 | 54.17 ± 0.45 |
| | Random Forest | 55.79 ± 0.61 | 49.07 ± 0.65 |
| | XGBoost | ***62.44 ± 0.96*** | ***59.22 ± 1.03*** |
| LW, RW, LF | Extra Trees | 52.46 ± 0.76 | 45.07 ± 0.84 |
| | LightGBM | 57.66 ± 1.21 | 53.73 ± 1.02 |
| | Random Forest | 55.83 ± 0.38 | 49.37 ± 0.21 |
| | XGBoost | **62.01 ± 1.18** | **58.82 ± 1.19** |
| LW, RW, RF | Extra Trees | 52.30 ± 0.46 | 45.28 ± 0.36 |
| | LightGBM | 57.57 ± 0.78 | 53.52 ± 0.56 |
| | Random Forest | 55.87 ± 0.52 | 49.23 ± 0.41 |
| | XGBoost | **62.31 ± 1.28** | **59.01 ± 1.32** |
| LW, ST, LF | Extra Trees | 49.82 ± 0.58 | 43.54 ± 0.56 |
| | LightGBM | 57.10 ± 0.89 | 52.89 ± 0.85 |
| | Random Forest | 54.58 ± 0.44 | 48.05 ± 0.33 |
| | XGBoost | **61.29 ± 0.80** | **57.74 ± 0.69** |
| LW, ST, RF | Extra Trees | 51.33 ± 0.42 | 42.88 ± 0.58 |
| | LightGBM | 57.46 ± 0.61 | 53.23 ± 0.44 |
| | Random Forest | 54.85 ± 0.44 | 48.44 ± 0.57 |
| | XGBoost | **61.73 ± 0.73** | **57.99 ± 0.67** |
| LW, LF, RF | Extra Trees | 48.93 ± 0.43 | 41.96 ± 0.68 |
| | LightGBM | 56.61 ± 1.06 | 52.01 ± 1.13 |
| | Random Forest | 53.84 ± 0.79 | 47.25 ± 0.76 |
| | XGBoost | **60.18 ± 0.90** | **56.22 ± 0.89** |
| RW, ST, LF | Extra Trees | 49.74 ± 0.57 | 41.95 ± 0.48 |
| | LightGBM | 57.16 ± 1.02 | 53.24 ± 0.95 |
| | Random Forest | 53.90 ± 0.94 | 46.59 ± 1.08 |
| | XGBoost | **60.88 ± 0.45** | **57.47 ± 0.51** |
| RW, ST, RF | Extra Trees | 50.37 ± 0.78 | 42.64 ± 0.89 |
| | LightGBM | 56.78 ± 0.61 | 52.88 ± 0.68 |
| | Random Forest | 52.08 ± 0.33 | 46.87 ± 0.57 |
| | XGBoost | **60.95 ± 0.81** | **57.58 ± 0.82** |
| RW, LF, RF | Extra Trees | 47.91 ± 0.75 | 41.32 ± 0.84 |
| | LightGBM | 55.66 ± 0.78 | 51.78 ± 0.69 |
| | Random Forest | 52.25 ± 0.50 | 45.78 ± 0.42 |
| | XGBoost | **60.03 ± 0.84** | **56.43 ± 0.85** |
| ST, LF, RF | Extra Trees | 43.90 ± 0.67 | 38.71 ± 0.61 |
| | LightGBM | 53.48 ± 0.66 | 48.46 ± 0.72 |
| | Random Forest | 49.98 ± 0.53 | 42.13 ± 0.60 |
| | XGBoost | **56.47 ± 0.37** | **51.89 ± 0.34** |

TABLE V
5-FOLD VCV RESULTS WITH FOUR-SENSOR COMBINATIONS

| Sensors | Model | Accuracy | $F1_{weighted}$ |
|---|---|---|---|
| LW, RW, ST, LF | Extra Trees | 51.17 ± 0.68 | 44.66 ± 0.78 |
| | LightGBM | 58.85 ± 1.19 | 55.34 ± 1.17 |
| | Random Forest | 56.20 ± 0.56 | 49.55 ± 0.53 |
| | XGBoost | **63.25 ± 0.67** | **60.23 ± 0.62** |
| LW, RW, ST, RF | Extra Trees | 52.66 ± 0.55 | 45.71 ± 0.48 |
| | LightGBM | 58.92 ± 1.13 | 55.08 ± 1.14 |
| | Random Forest | 55.95 ± 0.95 | 50.04 ± 0.93 |
| | XGBoost | ***63.57 ± 1.05*** | ***60.56 ± 0.99*** |
| LW, RW, LF, RF | Extra Trees | 51.47 ± 0.56 | 45.24 ± 0.40 |
| | LightGBM | 58.24 ± 1.17 | 54.45 ± 1.08 |
| | Random Forest | 55.84 ± 0.71 | 49.29 ± 0.59 |
| | XGBoost | **62.88 ± 1.08** | **59.69 ± 1.18** |
| LW, ST, LF, RF | Extra Trees | 51.37 ± 0.86 | 43.89 ± 0.98 |
| | LightGBM | 57.56 ± 0.86 | 53.31 ± 0.83 |
| | Random Forest | 54.93 ± 0.44 | 48.46 ± 0.41 |
| | XGBoost | **61.87 ± 0.58** | **58.32 ± 0.57** |
| RW, ST, LF, RF | Extra Trees | 49.89 ± 0.17 | 42.43 ± 0.24 |
| | LightGBM | 56.84 ± 0.62 | 53.08 ± 0.57 |
| | Random Forest | 51.59 ± 0.32 | 46.38 ± 0.28 |
| | XGBoost | **61.48 ± 0.71** | **58.32 ± 0.67** |

TABLE VI
5-FOLD VCV RESULTS WITH ALL FIVE SENSORS

| Sensors | Model | Accuracy | $F1_{weighted}$ |
|---|---|---|---|
| LW, RW, ST, LF, RF | Extra Trees | 51.57 ± 0.74 | 46.11 ± 0.76 |
| | LightGBM | 58.93 ± 0.95 | 55.22 ± 0.77 |
| | Random Forest | 56.41 ± 0.54 | 49.70 ± 0.59 |
| | XGBoost | ***64.01 ± 1.35*** | ***61.21 ± 1.38*** |

whole-body postural dynamics and improves discrimination between activities involving torso movements.

With three sensors (TABLE IV), the optimal configuration was (LW, RW, ST), again confirming the critical contribution of both wrists while demonstrating the added value of the pelvis. This combination achieved the highest F1-score of 59.22 ± 1.03 % using XGBoost, outperforming other triads such as those involving ankle sensors. The result reinforces the observation by Konak *et al.* [4] that the wrists are indispensable for capturing detailed activity patterns, while ST further consolidates recognition performance.

Adding a fourth sensor (TABLE V) led to further gains. The configuration (LW, RW, ST, RF) achieved 60.56 ± 0.99 % with XGBoost, indicating that ankle sensors contribute additional contextual information about locomotion and gross body movement. However, the magnitude of improvement compared with the three-sensor setup was modest (1.34 points).

Finally, the five-sensor configuration (TABLE VI) achieved the best absolute performance, with XGBoost reaching 61.21 ± 1.38 %. Nevertheless, the improvement from four to five sensors was marginal (0.65 points) and within the range of standard deviation. This finding mirrors the diminishing returns noted by Konak *et al.* [4], where the inclusion of both ankles (LF and RF) produced performance gains, but with a reduced effect compared with earlier increments.

Taken together, these results indicate a clear hierarchy of sensor contribution: the wrists are indispensable, the pelvis provides complementary postural context, and the ankles contribute marginal but non-negligible gains. From a practical perspective, the (LW, RW, ST) combination achieves nearly 97 % of the performance of the full five-sensor configuration while substantially reducing hardware and deployment overhead. This suggests that a three-sensor setup may represent an effective trade-off between accuracy and system complexity in operational nursing settings.

### *B. Optimal Model Configurations and Validation*

TABLE VII summarizes the best-performing configurations of the three ensemble models under the strict memory constraint. A consistent pattern emerges across all models: the inclusion of both wrist sensors (LW and RW) remains indispensable, while the addition of the pelvis (ST) and ankle (LF or RF) sensors provides complementary postural and locomotor information.

Among the models, XGBoost achieved the highest overall performance, reaching an NCV weighted F1-score of 61.21 ± 1.38 % with the full five-sensor setup (LW, RW, ST, LF, RF). Despite its superior accuracy, this model incurred

TABLE VII

MODEL CONFIGURATIONS UNDER THE 1 MB MEMORY CONSTRAINT, WITH PARAMETERS, NCV RESULTS, AND MODEL STATISTICS

| Model | Sensor subset | Configuration† | VCV (Hyperparameters) | NCV $F1_{weighted}$ (%) | Model size (MB) | Training time (s) | Prediction time (s) |
|---|---|---|---|---|---|---|---|
| Extra Trees§ | LW, RW, ST, LF, RF | n_estimators ∈ [1, 5]<br>max_depth ∈ [1, 12] | n_estimators = 3<br>max_depth = 11 | 45.65 ± 1.07 | 0.986 | **0.1784** | **0.0118** |
| LightGBM§ | LW, RW, ST, LF | n_estimators ∈ [1, 9]<br>max_depth ∈ [1, 10] | n_estimators = 9<br>max_depth = 7 | 55.34 ± 1.17 | 0.995 | 3.5015 | 0.0175 |
| Random Forest‡ | LW, RW, ST, RF | n_estimators ∈ [1, 9]<br>max_depth ∈ [1, 11] | n_estimators = 6<br>max_depth = 10 | 50.04 ± 0.93 | 0.987 | 2.3947 | 0.0119 |
| XGBoost‡ | LW, RW, ST, LF, RF | n_estimators ∈ [1, 29]<br>max_depth ∈ [1, 3] | n_estimators = 28<br>max_depth = 3 | **61.21 ± 1.38** | **0.975** | 36.4894 | 0.0213 |

†Hyperparameter configuration includes the number of trees and tree depth, with `random_state=42`; all other parameters were kept at their default settings. Model conversion to embedded C header files: §`m2cgen`, ‡`micromlgen`.

the longest training time (36.4894 s), reflecting the computational cost of gradient boosting. Nevertheless, prediction time remained short (0.0213 s), and the model size (0.975 MB) was the smallest among all classifiers.

LightGBM offered a competitive balance between accuracy and efficiency, attaining 55.34 ± 1.17 % with four sensors (LW, RW, ST, LF). Its model size (0.995 MB) approached the imposed memory ceiling, but training (3.5015 s) and inference (0.0175 s) were markedly faster than XGBoost. This efficiency, together with its histogram-based optimization, makes LightGBM a practical choice when rapid retraining and lower computational cost are prioritized.

Random Forest achieved moderate predictive performance (50.04 ± 0.93 %) under its best configuration (LW, RW, ST, RF). With a compact model size (0.987 MB), it required 2.3947 s for training and 0.0119 s for inference, demonstrating balanced runtime efficiency. Its bagging-based structure stabilizes learning across trees and reduces variance, yet it struggles to capture cross-sensor dependencies under tight memory constraints, leading to lower accuracy compared with the boosting-based models.

Extra Trees yielded the lowest predictive accuracy (45.65 ± 1.07 %) but was the fastest model overall. It trained in only 0.1784 s and inferred in 0.0118 s, with a compact 0.986 MB model size. The randomized feature-splitting mechanism accelerates computation and lowers variance, but limits its ability to model complex inter-sensor relations. Despite this, its minimal latency and small footprint make Extra Trees suitable for ultra-lightweight, low-power implementations for edge deployment where inference speed is critical.

### C. Error Analysis and Class-Level Performance

Fig. 2 and TABLE VIII provide a detailed breakdown of the class-level performance of the best XGBoost model trained with all five sensors. The overall weighted F1-score of 61.21 % corresponds to moderate recognition ability, but a closer inspection reveals substantial variability across activities.

High-performing classes include Blow-dry (94.92 % F1) and Put medication (88.74 % F1), which exhibit distinctive hand motions and are well represented in the training data. Similarly, frequent daily activities such as Null activity, Serve food, and Put food on plate achieved balanced precision and recall above 70 %, indicating that the system can reliably detect common and kinematically salient behaviors.

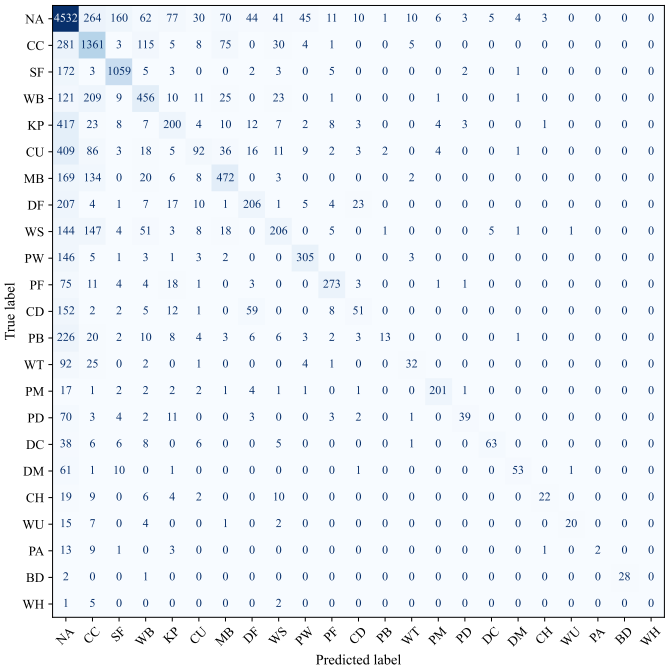


Fig. 2. Confusion matrix of the XGBoost using all five sensors

In contrast, several fine-grained or infrequent activities remain highly challenging. For example, Prepare bath and Put accessories achieved F1-scores below 15 %, largely due to severe class imbalance (TABLE I) and their strong overlap with semantically similar tasks in the feature space. The complete failure to recognize Wash hair (no true positives) reflects the scarcity of training samples (only 8 windows), underscoring the limitations of data-driven models under extreme imbalance. Activities involving locomotion or mixed upper–lower body coordination, such as Clean up (20.72 %) and Wheelchair transfer (30.33 %), also suffered, likely because transitional movements are easily confused with neighboring classes in the temporal windowing.

TABLE VIII
CLASSIFICATION REPORT OF THE XGBOOST USING ALL FIVE SENSORS

| Activity | Precision (%) | Recall (%) | F1-score (%) |
|---|---|---|---|
| Null activity | 61.42 | 84.27 | 71.05 |
| Change clothes | 58.29 | 72.09 | 64.46 |
| Serve food | 82.80 | 84.38 | 83.58 |
| Wash in bed | 57.87 | 52.60 | 55.11 |
| Kitchen preparation | 51.81 | 28.21 | 36.53 |
| Clean up | 48.17 | 13.20 | 20.72 |
| Make bed | 66.11 | 57.99 | 61.78 |
| Deliver food | 58.03 | 42.39 | 48.99 |
| Wash at sink | 58.69 | 34.68 | 43.60 |
| Push wheelchair | 80.69 | 65.03 | 72.02 |
| Put food on plate | 84.26 | 69.29 | 76.04 |
| Collect dishes | 51.00 | 17.47 | 26.02 |
| Prepare bath | 76.47 | 4.23 | 8.02 |
| Wheelchair transfer | 59.26 | 20.38 | 30.33 |
| Put medication | 92.63 | 85.17 | 88.74 |
| Pour drinks | 79.59 | 28.26 | 41.71 |
| Dental care | 86.30 | 47.37 | 61.17 |
| Documentation | 85.48 | 41.41 | 55.79 |
| Comb hair | 81.48 | 30.56 | 44.44 |
| Wipe up | 90.91 | 40.82 | 56.34 |
| Put accessories | 100 | 6.90 | 12.90 |
| Blow-dry | 100 | 90.32 | 94.92 |
| Wash hair | 0 | 0 | 0 |

## IV. CONCLUSION

This study provided a systematic evaluation of classical ensemble learning methods for nursing care activity recognition on the SONAR dataset under strict memory constraints relevant to LPMC, offering a deployment-oriented benchmark for resource-efficient modeling. The results indicate that boosting-based approaches, particularly XGBoost, achieve the strongest overall performance, while analysis of individual and combined sensor placements highlights the consistent importance of wrist sensors and the complementary contributions of the pelvis and ankles.

Despite these findings, several limitations remain: the SONAR dataset contains substantial missing values and pronounced class imbalance, the exclusion of NaN-affected segments reduces data volume, and generalization to unseen users or clinical environments requires further validation. Future work should explore imputation–deletion ablation analyses, cost-sensitive learning and data augmentation for rare activities, lightweight model compression, and empirical validation on embedded hardware to better assess latency and energy profiles.

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